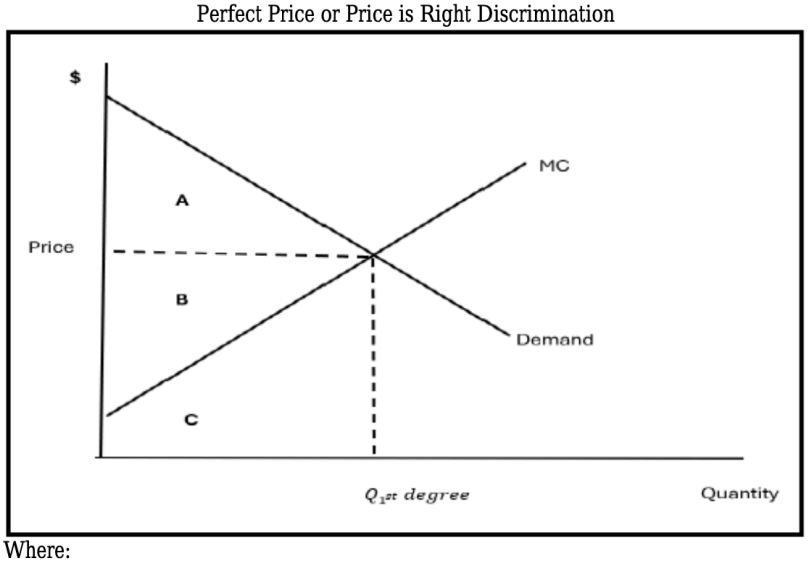
ECONOMICS Q2. Technology now allows personalised pricing. If this came to be widely used, what effects should we expect?

Things have changed; everything has changed since the introduction of technology; the world will change day by day. Advances in Big Data, Artificial Intelligence (AI), and online platforms have driven the shift from a uniform "price-tag" model towards personalised pricing, where prices can be tailored to how much customers are prepared to spend. By collecting information on consumers' age, location, search history, and purchasing behaviour, firms can estimate an individual's willingness to pay with higher precision, allowing businesses to move beyond broad customer groups towards pricing each consumer differently.

Indeed, technological development has greatly reduced the time and cost of collecting customer data through automation, thanks to Big Data and online platforms. A business could charge differentiated prices to different customer segments on ride-hailing services (Grab) and e-commerce platforms (Shopee). Although current technologies have brought economic benefits, they remain imperfect because algorithmic limitations and privacy concerns prevent the efficient implementation of first-degree price discrimination. As personalised pricing grows in popularity, its effects will extend beyond greater efficiency to include questions of fairness, market power, and consumer autonomy.

To understand the effects of personalised pricing, it is first necessary to examine the theory of first-degree price discrimination. According to Yuqi Wang, first-degree (or perfect) price discrimination occurs when a monopoly charges each customer a different price based on their maximum willingness to pay. Under this model, consumer surplus is theoretically eliminated because each customer pays exactly what they are willing to pay.

Perfect Price or Price is Right Discrimination



Where:
A = Consumer Surplus
B = Producer Surplus
C = Opportunity Cost
A + B = Area of first-degree price discrimination
Author's Creation, 2022.

Figure 1: Perfect price discrimination or First-degree price discrimination in A Simplified Theoretical Understanding of Price Discrimination as a Business Management Strategy by Ejiro U. Osiobe

In this diagram, consumer surplus (Area A) represents the economic benefit consumers gain, lying below the demand curve and above the market price. Producer surplus (Area B), on the other hand, represents the producer's benefit, lying above the marginal cost (MC) curve and below the market price. Under first-degree price discrimination, Area A is transferred to producers, so producer surplus expands to Area A+B, while consumer surplus is reduced to zero. Although this improves allocative efficiency by eliminating deadweight loss, it also raises concerns about fairness. If consumers perceive that they are being treated unfairly, their willingness to purchase may decline.

An early example of dynamic pricing appeared in 1999 when Coca-Cola's CEO, Douglas Ivester, proposed vending machines that would automatically raise prices as outdoor temperatures rose. The proposal immediately triggered public criticism because consumers perceived it as exploiting their immediate needs rather than responding to normal market competition. This illustrates that while personalised pricing may enhance economic efficiency, consumers often judge prices not only by efficiency but also by their perceived fairness.

Grab and Shopee provide useful examples of how personalised pricing operates in digital markets. As two-sided platforms, they connect consumers with drivers or sellers while collecting large amounts of behavioural data through users' transactions and online activities. They then estimate customers' willingness to pay and offer different prices, discounts, or

vouchers to different users, and that's the way technological advances have made personalised pricing feasible in practice.

I will use Grab and Shopee as examples to illustrate the positive effects of personalised pricing. Grab uses GPS data, route distance, traffic, and weather conditions to estimate fares. At the same time, Shopee offers different vouchers based on users' purchase history, shopping behaviour, engagement with platform features, and fraud detection systems. This is how digital platforms tailor prices and discounts to individual users.

In the digital era, consumers increasingly value convenience and easy access to goods and services. As large digital platforms, Grab and Shopee connect millions of users with drivers and sellers through automated two-sided markets. Their widespread use means that firms can collect vast amounts of behavioural data and make personalised pricing more accurate. According to Tuoi Tre, ShopeeFood accounted for 56% of Vietnam's online food delivery market during the seven days in April 2025. The larger the platform, the richer the data available for estimating the maximum price consumers would accept.

Additionally, the scale of platforms such as Grab and Shopee strengthens the effectiveness of personalised pricing. By integrating multiple services, including transport, food delivery, logistics, and e-commerce, these platforms benefit from economies of scale and accumulate even more consumer data across different activities. As Zexiang Wang argues, integrating a wide range of services allows Grab to share infrastructure and reduce operational costs. Similarly, interactive features such as Shopee Live and gamified rewards encourage users to spend more time on the platform, generating richer behavioural data that boosts the accuracy of pricing algorithms.

The convenience, trust, and large user base of platforms such as Grab and Shopee provide favourable conditions for Big Data and Artificial Intelligence to collect behavioural data and estimate personalised prices more accurately. As pricing becomes tailored to individual consumers, firms are able to allocate goods and services more efficiently and reap potential economic benefits.

According to the OECD (2018), the data collected by digital platforms can be broadly divided into three categories: data inferred from consumer behaviour, data voluntarily provided by users, and data directly observed by firms, such as cookies. Together, this information enables companies to set prices that better align with what each customer is willing to pay.

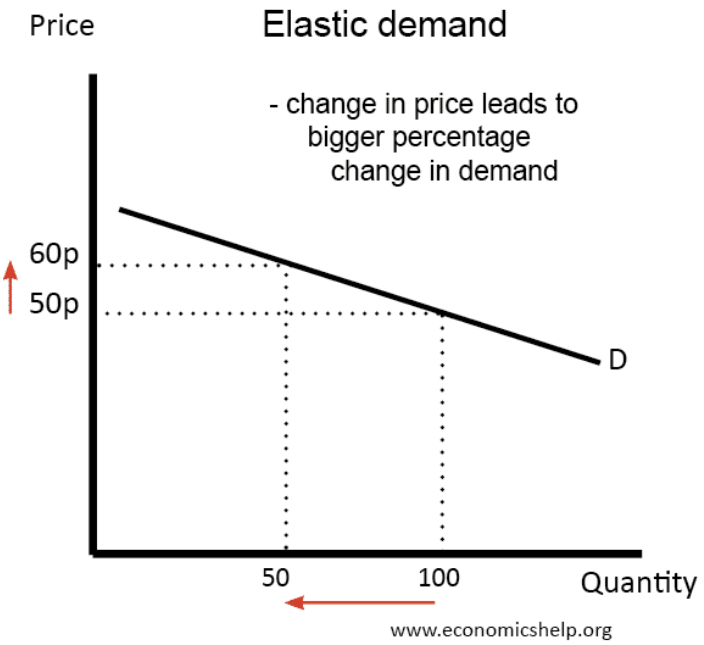


Figure 2: Price Elastic Demand by Economics Helps

As more people use Grab and Shopee, vast amounts of consumer data will be generated. Artificial Intelligence will utilize algorithms to calculate the price elasticity of demand for each customer. Instead of the uniform price tag, the algorithms would calculate the Shopee voucher and Grab's price subsidy, and firms can attract consumers who would otherwise be unwilling or unable to purchase at a uniform price. Consumers with relatively inelastic demand, such as higher-income users, are more likely to purchase without discounts, whereas price-sensitive consumers, including many students, respond more strongly to vouchers and subsidies. By estimating these differences in price elasticity, firms can offer targeted discounts that stimulate additional demand. As a result, personalised pricing may reduce deadweight loss by enabling transactions that would not occur under a single uniform price.

For instance, when there appears to be a bad weather day or in rush hour, Grab would automatically raise the price: this would encourage drivers to accept the trip to delete scarcity, achieving the market equilibrium and completely solving the deadweight loss. Indeed, this is a market signal illustrating dynamic pricing on the law of supply and demand.

Moreover, by integrating multiple services, platforms such as Grab and Shopee may benefit from economies of scale and cross-subsidisation. Revenue gained from premium services, advertisements, or financial products can help fund discounts and vouchers for more price-sensitive consumers. In this way, personalised pricing has the potential to expand access while improving overall market efficiency. Likewise, free shipping vouchers also encourage additional purchases by lowering consumers' effective prices, thereby raising demand on the platform.

However, personalised pricing on platforms such as Grab and Shopee also presents economic, ethical, and privacy concerns. Because personalised pricing relies heavily on

predictive algorithms, consumers remain vulnerable to algorithmic errors and information asymmetry. As Sandra Wachter notes, one limitation of AI is the “black box” problem. Algorithms may become overfitted to historical data, making them less capable of responding to unprecedented events or unusual consumer behaviour.

Ethical issues can be observed on platforms such as Grab, where a consumer who once accepted an unusually high fare during an emergency may subsequently be identified as having a relatively inelastic demand. Thus, future prices offered to that consumer may remain higher than their actual willingness to pay. In this case, an exceptional purchase is mistaken for a stable pattern of consumer behaviour. Such information asymmetry may gradually reduce consumers’ autonomy. According to Shoshana Zuboff, firms that possess extensive behavioural data gain unprecedented power to influence consumer decisions, turning personal information into a source of economic control.

From a Lockean perspective (Second Treatise of Government), this systemic rigidity reveals a moral concern surrounding technological monopolies: individuals may increasingly be priced according to algorithmic predictions derived from their past behaviour. In Venditio (1695), Locke points out that while the just price of a good is determined by the market, exploiting an individual’s momentary distress to demand an excessive price undermines the moral legitimacy of a transaction.

From an economic perspective, firms with substantial market power may use the data they collect to charge prices closer to each consumer’s willingness to pay. When personalised pricing becomes more sophisticated, consumer surplus may gradually decline as a larger share of the gains from trade is transferred to producers. Such pricing increases the risk that market power will be reinforced through information advantages instead of fair competition.

Continuing with privacy concerns, personalised pricing inevitably depends on the large-scale collection of personal data. Many consumers have grown concerned about how platforms track their online activities through browsing histories and cookies. Consequently, users may become less willing to share personal information, and trust in digital platforms would weaken.

In contrast, if customers happen to find themselves being taken advantage of or treated unfairly, they would implement their own strategic customer behavior (repeated game theory) as an act of proactive resistance: turning off cookies, locking in a clone account, or quitting the app. Autonomy does not completely disappear, but it transforms into gaming the system, illustrating that their demand is inelastic. According to Paul Klemperer in Oxford University, a switching cost results from a customer’s desire for compability between his current purchase and a previous investment. In the situation of Grab’s purposes to maintain high costs beyond their abilities to pay for, customers would churn Grab, and switch to other platforms such as Gojek, Be, and GreenSM (in Vietnam). This would cause the economic loss in Grab, and algorithms had to self-corrrect to maximize the frequencies of transaction.

On the other hand, if low-income consumers are charged disproportionately high prices for essential goods such as medicine because of information asymmetry, they may delay purchasing them or switch to lower-quality alternatives. In this sense, personalised pricing risks limiting access to essential goods for the most vulnerable consumers, violating ethical principles of market transactions established by Locke. Nevertheless, the above-mentioned cross-subsidization would solve this issue. Instead of implementing uniform pricing, dynamic pricing assists in more closely approaching those who have low incomes. Instead of this strategy, the poor would be the first to be left out because those prices are beyond their means, violating autonomy through the Lockean lens.

To solve these issues, Grab and Shopee should implement regulatory boundaries, algorithmic transparency, and price caps. To solve issues related to regulatory boundaries, there should be a regulation related to opt-in and opt-out. By definition (GDPR), opt-in means the individual must take an affirmative action before processing begins, while opt-out means that if processing proceeds by fault, the individual has the right to object or withdraw at a later point. As a result, Grab should delete and reset the data after the emergency cases (rush hour, low battery level, and bad weather) to ensure regulatory boundaries. Furthermore, algorithms and Artificial Intelligence features (OECD, 2018) should be improved to maximize data privacy and solve the “Black Box” of Artificial Intelligence. Related to essential products, especially medicines, authorities should examine and control automated pricing to provide each person with these supplies because this is included in their right to life.

In conclusion, with advances in Big Data, Artificial Intelligence, and digital platforms, ride-hailing and e-commerce services have moved closer to first-degree price discrimination, making personalised pricing more feasible than ever. Although personalised pricing can achieve better economic efficiency by reducing information costs and expanding access for some consumers, its widespread adoption also raises ethical concerns over algorithmic bias and information asymmetry, as well as economic and privacy challenges. Ultimately, the effects of personalised pricing will depend on whether technological innovation can improve market efficiency without sacrificing fairness, consumer trust, and transparency.

REFERENCE LIST

1. The Analysis of Grab's Business Model and Revenue Strategies – Zeyu Wu
https://www.researchgate.net/publication/387716884_The_Analysis_of_Grab's_Business_Model_and_Revenue_Strategies

2. Price Discrimination in the Era of Big Data – Yuqi Wang
https://www.researchgate.net/publication/363029282_Price_Discrimination_in_the_Era_of_Big_Data

3. Privacy, Economics, and Price Discrimination on the Internet – Andrew Odlyzko
http://archive.dimacs.rutgers.edu/Workshops/AuctionDesign/slides/privacy.economics.pdf

4. First-Degree Price Discrimination Using Big Data – Benjamin Reed Shiller
https://www.brandeis.edu/economics/RePEc/brd/doc/Brandeis_WP58R2.pdf

5. Personalised Pricing in the Digital Era – OECD Roundtables on Competition Policy Papers
https://www.oecd.org/en/publications/personalised-pricing-in-the-digital-era_db4d9c9c-en.html

6. Big Data and Personalized Pricing – Etye Steinberg
https://www.jstor.org/stable/26867943

7. Coke Tests Vending Unit That Can Hike Prices in Hot Weather – Constance L. Hays
https://archive.nytimes.com/www.nytimes.com/learning/students/pop/articles/coke-vending.html

8. A Simplified Theoretical Understanding of Price Discrimination as a Business Management Strategy – Ejiro U. Osiobe
https://www.researchgate.net/publication/361284590_A_Simplified_Theoretical_Understanding_of_Price_Discrimination_as_a_Business_Management_Strategy

9. Real-Time Spatial–Intertemporal Pricing and Relocation in a Ride-Hailing Network: Near-Optimal Policies and the Value of Dynamic Pricing
https://pubsonline.informs.org/doi/abs/10.1287/opre.2022.2425

10. On Social Contract Theory – Acilyn Aspiras
https://www.researchgate.net/publication/353163959_On_Social_Contract_Theory

11. Social Contract Theory
https://niu.edu.in/sla/online-classes/Social-contract-theory.pdf

12. The effects of surge pricing on driver behavior in the ride‐sharing market: Evidence from a quasi‐experiment

https://www.researchgate.net/publication/364114507_The_effects_of_surge_pricing_on_driver_behavior_in_the_ride-sharing_market_Evidence_from_a_quasi-experiment

13. Dynamic Pricing Risks Eroding Consumer Trust: Gartner – Samantha Nelson
https://consumergoods.com/dynamic-pricing-risks-eroding-consumer-trust-gartner

14. AI-Driven Dynamic Pricing: Erosion of Consumer Welfare, Invisible Hand, and Rise of Platform Quasi-Taxation – Sang Yop Kang
https://www.cambridge.org/core/journals/german-law-journal/article/aidriven-dynamic-pricing-erosion-of-consumer-welfare-invisible-hand-and-rise-of-platform-quasitaxation/427EC3E1A84ED4C77A772FCB94BE48CB

15. Dynamic Pricing and Elasticity – Jessie Martinez
https://mlpp.pressbooks.pub/microeconomicscaseissues/chapter/dynamic-pricing-and-elasticity/

16. How Grab works – Anuraag Jain
https://oyelabs.com/how-grab-works-business-model-explained/

17. What Is Shopee and How Does It Work? – Abhinav Saini
https://miracuves.com/blog/what-is-shopee-and-how-it-work/

18. ShopeeFood tops Vietnam’s online food delivery orders in survey, Grab leads by market share
https://news.tuoitre.vn/shopeefood-tops-vietnams-online-food-delivery-orders-in-survey-grab-leads-by-market-share-103250709133722885.htm

19. The Strategic Analysis of Grab's Strategic Dominance in Southeast Asia – Zexiang Wang
https://share.google/X2VLxhRjEK5X1pHhY

20. Shopee Market Insights: An In-depth Market Analysis, Growth, and Competition
https://cube.asia/read/shopee-market-insights-an-in-depth-market-analysis-growth-and-competition/

21. Grab/Uber merger: CCCS Provisionally Finds that the Merger Has Substantially Lessened Competition, Proposes Directions to Restore Market Contestability and to Impose Financial Penalties
https://www.ccs.gov.sg/media-and-events/newsroom/announcements-and-media-releases/grab-uber-merger--cccs-provisionally-finds-that-the-merger-has-substantially-lessened-competition--proposes-directions-to-restore-market-contestability-and-to-impose-financial-penalties/

22. Dynamics of Business Competition in the Business Sector Online Transportation: Case Study of Grab Indonesia's Discriminatory Allegations Against Independent Drivers According to the Business Competition Supervisory Commission's Perspective

https://ejournal.mandalanursa.org › viewFile

23. Repeated Game
https://gfrechette.com › print

24. Competition when Customers have Switching Costs - An Overview with Applications to Industrial Organization_Macroeconomics and International Trade
https://www.researchgate.net/publication/4995480_Competition_When_Consumers_Have_Switching_Costs_An_Overview_with_Applications_to_Industrial_Organization_Macroeconomics_and_International_Trade

25. Opt-in Opt-out GDPR Guide
https://www.legiscope.com/blog/opt-in-opt-out-gdpr-guide.html